

API ENDPOINT PLAN

HTTP Method	Route	Description	Role Required	Request Body	Expected Response
POST	/api/auth/register	Registers a new user account as an Organiser or Participant.	None (Public)	{ "fullName": "string", "email": "string", "password": "string", "role": "string" }	201 Created: {"userId": 1, "email": "...", "role": "..."} 400 Bad Request: Validation failure 409 Conflict: Email already registered
POST	/api/auth/login	Authenticates user credentials and establishes server-side session state.	None (Public)	{ "email": "string", "password": "string" }	200 OK: {"userId": 1, "fullName": "...", "role": "..."} 400 Bad Request: Missing credentials 401 Unauthorized: Invalid email or password
POST	/api/auth/logout	Invalidate and terminate the active session.	Any (Authenticated)	<i>None</i>	200 OK: {"message": "Logged out successfully"} 401 Unauthorized: Session

					expired or absent
GET	/api/users/profile	Retrieves current authenticated profile attributes.	Any (Authenticated)	None	200 OK: User profile JSON 401 Unauthorized: Not authenticated 404 Not Found: User record not found
PUT	/api/users/profile	Updates profile attributes and profile image URL.	Any (Authenticated)	{ "fullName": "string", "profilePicUrl": "string" }	200 OK: Updated user profile JSON 400 Bad Request: Validation failure 401 Unauthorized: Not authenticated
GET	/api/events	Retrieves all available road running, walking, and cycling events.	None (Public)	None	200 OK: Array of Event objects with category summaries
GET	/api/events/{id}	Retrieves detailed metadata and categorie	None (Public)	None	200 OK: Event details JSON

		s for an event.			404 Not Found: Event does not exist
POST	/api/events	Creates a new event record.	Organiser	<pre>{ "eventName": "string", "description": "string", "eventDate": "YYYY-MM-DD", "location": "string", "distanceKm": 10.0, "eventType": "Run", "bannerUrl": "string" }</pre>	201 Created: Created Event JSON 400 Bad Request: Missing or invalid fields 401 Unauthorized: Unauthenticated 403 Forbidden: Participant role
PUT	/api/events/{id}	Replaces or updates existing event information.	Organiser	<pre>{ "eventName": "string", "description": "string", "eventDate": "YYYY-MM-DD", "location": "string", "distanceKm": 10.0, "eventType": "Run", "bannerUrl": "string" }</pre>	200 OK: Updated Event JSON 400 Bad Request: Route ID mismatch or bad input 403 Forbidden: User does not own event 404 Not Found: Event not found

DELETE	/api/events/{id}	Deletes an event and its dependent sub-resources .	Organiser	None	204 No Content: Resource removed successfully 401 Unauthorized: Unauthenticated 403 Forbidden: User does not own event 404 Not Found: Event not found
GET	/api/events/{eventId}/categories	Retrieves all sub-categories configured for an event.	None (Public)	None	200 OK: Array of Category objects 404 Not Found: Event does not exist
POST	/api/events/{eventId}/categories	Appends a distance or age division to an event.	Organiser	{ "categoryName": "string", "description": "string" }	201 Created: Created Category JSON 400 Bad Request: Invalid input 401 Unauthorized:

					Unauthenticated 403 Forbidden: Participant role 404 Not Found: Event not found
POST	/api/enrolments	Enrols the logged-in participant into an event category.	Participant	{ "eventId": 1, "categoryId": 2 }	201 Created: Enrolment record JSON 400 Bad Request: Category does not belong to Event 401 Unauthorized: Unauthenticated 403 Forbidden: Organiser role 409 Conflict: Participant already enrolled
GET	/api/enrolments/my-enrolments	Lists all race registrations	Participant	None	200 OK: Array of Enrolment

		ns belonging to the current participan t.			records with nested Event summaries 401 Unauthorized: Unauthentic ated 403 Forbidden: Organiser role
GET	/api/events/{eventId}/enrolments	Lists all participan t entries for an organiser' s event.	Organiser	None	200 OK: Array of enrolled participants with selected categories 401 Unauthorized: Unauthentic ated 403 Forbidden: User does not own event 404 Not Found: Event not found
POST	/api/results	Records official finish times and overall positions.	Organiser	{"enrolmentId": 1, "participantId": 3, "finishTime": "01:45:22", "position":	201 Created: Captured Result JSON

				12, "totalFinishers": 150}	400 Bad Request: Position exceeds total finishers 401 Unauthorized: Unauthenticated 403 Forbidden: Participant role 409 Conflict: Result already recorded for enrolment
GET	/api/results/my-results	Retrieves performance history across past events for the caller.	Participant	None	200 OK: Array of personal result records 401 Unauthorized: Unauthenticated 403 Forbidden: Organiser role
POST	/api/media/upload	Uploads an image file to	Any (Authenticated)	multipart/form-data	200 OK: { "blobUrl": "https://..." }

		Azure Blob Storage and returns its URI.		(Binary file payload)	400 Bad Request: Unsupported media type or empty file 401 Unauthorized: Unauthenticated
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Reference List

Microsoft, 2023. RESTful web API design - Best practices for cloud applications.

Microsoft Learn: Azure Architecture Center. Available at:

<https://learn.microsoft.com/en-us/azure/architecture/best-practices/api-design>.